

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5108
Subject Title	Future Air Mobility and Low-altitude Airspace Management
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. an overview and introduction to next generation air mobility systems, mainly including Urban Air Mobility (UAM) and green aviation. 2. the latest research and technological advancements in future air mobility. 3. the skills to design and implement air traffic management and vertiport design and operations in air mobility system.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand and apply the principles of aviation operations in the context of future air mobility. b. a critical analysis of the current issues and challenges for development of UAM, such as safety, noise, etc. c. identify and solve emerging complex problems related to UAM using optimization, learning, and control methods. d. design and implement solutions to enhance safety and efficiency for UAM.
Subject Synopsis/ Indicative Syllabus	Introduction: Overview and Basic Concepts in emerging UAM, including vehicle-based aspects, public acceptance of UAM, examples of Unmanned Aircraft Vehicles (UAV) application, prospects and drawbacks for introducing UAM, and potential solutions. Challenges and Impact Factors of UAM: The exploration of challenges includes questions on safety, noise, energy efficiency, environmental impact, certification and policy, as well as challenges in the area of traffic management and ground infrastructure requirements. Airspace structure: The definition, design, and categorization of low-altitude airspace for UAM. The operating environments include airspaces, types of operations, regulations, and procedures necessary to support an operation. Aircraft types and performance: The latest technological advancements in low-altitude aircrafts, including electric aircraft, drones and others.

	Aircraft categorization and performance. Introduce the regulatory requirements of the airspace that all aircraft are required to comply with within which they are conducting operations. UAM terminal: terminal area design to support UAM departure and arrival operations, operations and management for UAM, including vertiport/ vertistop location, design and management. UAM operators are expected to utilize whichever vertiport configuration meets their operational needs. Airfield capacity and delay: Low-altitude airfield capacity evaluation and delay management. Introduce demand-capacity balancing technique when the requested resources cannot support the collective UAM Operational Intent demand. Demand management: Demand evaluation and management strategies for congestion mitigation in UAM. Technical methods such as machine learning for demand forecasting. UAM traffic management: Air traffic management, air traffic control strategies for UAM. Introduce new ATC tactical deconfliction techniques, including the formulation of new separation standards that would rely on enhanced aircraft performance and air traffic management system fidelity may be utilized.																								
Teaching/Learning Methodology	Teaching is conducted through class lectures, quiz and group project. The basic concepts, research methodology and theoretical models will be introduced. The understanding problems and challenges in emerging UAM and solution strategies development will be emphasized. Research methodology, case study and analytics skills are taught in class as well as the related real-life scenarios to enhance the teaching and learning abilities. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Quiz</td><td>√</td><td></td><td>√</td><td>√</td></tr><tr><td>Group project</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Quiz	√		√	√	Group project	√	√	√	√
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Group project	√	√	√	√																					

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Assignment	20%	√	√	√	
	2. Quiz	5%	√	√	√	
	3. Group project	35%	√	√	√	√
	4.Final examination	40%	√	√	√	√
	Total	100%				
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.6 \times \text{Continuous Assessment} + 0.4 \times \text{Final Examination}$ The continuous assessment (60%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via assignment, group project and quiz. In particular, group project is used to assess the students’ capability of self-study and problem-solving and effective communication with team members so as to fulfil the requirements of working in the aviation industry. The quiz can help students involved more in understanding the application of principles from lectures in real problems. The final examination (40%) will also be considered to assess the students’ learning outcomes in mathematical problem formulation and solution design for questions in the context of UAM.						
Student Study Effort Expected	Class contact:					
	▪ Lecture/Case Study					39 Hrs.
	Other student study effort:					
	▪ Self-learning/preparation					36 Hrs.
	▪ Literature study/case study/reading					36 Hrs.
	Total student study effort					111 Hrs.
Reading List and References	1. Boyles, S. D., N. E. Lownes, and A. Unnikrishnan. (2023) Transportation Network Analysis, Volume I, Version 0.91. 2. Balakrishnan, R., Ranganathan K., A Textbook of Graph Theory. 2nd ed. 2012 Edition 3. L’Afflitto, A., Inalhan, G., Shin, H. Control of Autonomous Aerial Vehicles. 2024 Edition. 4. Bradley S. P. Hax A. C. & Magnanti T. L. (1977). Applied mathematical programming. Addison-Wesley Pub.					

	5. Arblaster, M. Air Traffic Management: Economics, Regulation and Governance. 2018 Edition. 6. FAA. (2023) Concepts of Operations: Unmanned Aircraft Systems Traffic Management.
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Dec 2023